EDUHUB / ENGINEERING DOCUMENTATION

DATA FLOW DIAGRAM

EduHub | Online Learning Platform

01	01	07
CONTEXT DFD	LEVEL 1 MODEL	LEVEL 2 VIEWS

Logical process model, balanced external interfaces, data-store mappings, detailed decomposition and review checklist.

Source: supplied EduHub project brief (5 pages). Previously prepared architecture, ERD, schema, use case and requirements documents inform continuity.

DFD model at a glance

Logical decomposition of the EduHub online learning platform

01	07	07	10
CONTEXT DIAGRAM	LEVEL 1 PROCESSES	LEVEL 2 EXPANSIONS	LOGICAL STORES

Purpose and boundary

This deliverable shows where information originates, which business process transforms it, where it is persisted, and which external party receives it. It supplements - rather than repeats - the use case, ERD, software architecture and API documents.

Included

Context (Level 0), platform Level 1, seven Level 2 decompositions, labeled flows, grouped logical stores, boundary balancing, a data dictionary and an audit checklist.

Not included

Sequence timing, system-state transitions, physical SQL keys, request/response API contracts, internal MVC classes or executable deployment wiring.

Consistency baseline

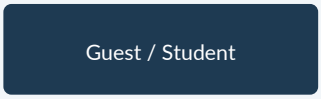
Primary source	EduHub five-page graduation-project specification: course authoring, Azure videos, Stripe, progress, quizzes, ratings, certificates, admin.
Companion models	N-Tier software architecture; 38-entity ERD and schema; 48 use cases; 83 functional and 32 quality requirements.
Model stance	Logical data movements with proposed cross-cutting safeguards; not proof of an existing production workflow.

Notation & rules

Gane-Sarson-inspired logical DFD notation; no UML sequence semantics

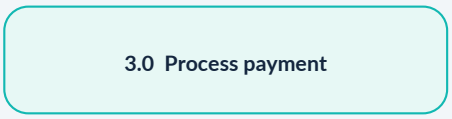
The four modeling elements

EXTERNAL ENTITY



A source or sink outside EduHub; dark rectangle.

PROCESS



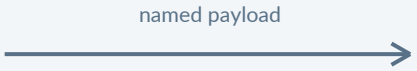
A transformation; numbered rounded rectangle.

DATA STORE



Logical persistence group; record-like rectangle.

DATA FLOW



Directed, named information movement.

DFD conventions used throughout

R1	Each flow is named for data content, not a button or control command.
R2	Data stores connect to processes only; no actor-to-store or store-to-store edges.
R3	All processes have meaningful data input and output, with a stable hierarchical ID.
R4	Context boundary inputs/outputs are conserved when exploded into Level 1.
R5	Level 2 refines a Level 1 process without adding unsupported external exchanges.
R6	Data stores are logical groups, not necessarily one database table or Azure container.

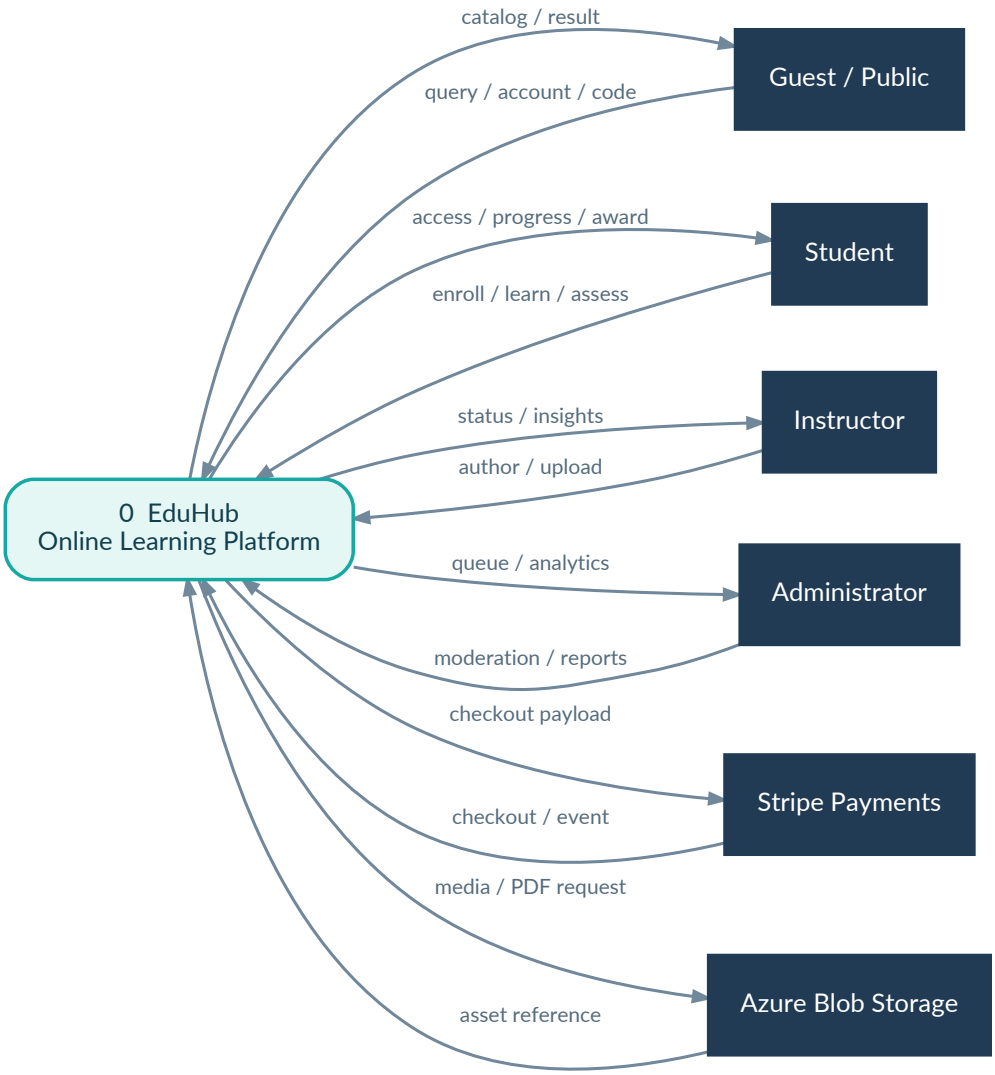
Interpretation

DFD flows show what data moves, not the execution order, elapsed time, HTTP mechanism or branching decisions. Use the companion sequence/activity diagrams for those concerns.

Context DFD / Level 0

One process models EduHub; all actor and provider exchanges cross its boundary

FIGURE 01 / SYSTEM CONTEXT



Boundary interpretation

Four human roles interact with one EduHub process: Guest/Public, Student, Instructor and Administrator. Stripe is the external checkout and payment-event system. Azure Blob Storage holds protected media and generated certificate artifacts; the database stores only their references.

> **No internal stores on the context page**
>
> D1-D10 are deliberately absent here. The Level 1 diagram decomposes process 0 into internal transformations and logical records.

Context flow inventory

Every external exchange has a direction, payload and Level 1 owner

Context interface balancing baseline

B01-B12

ID	DIRECTION	NAMED DATA CONTENT	LEVEL 1
B01	Guest -> EduHub	Search/filter parameters; registration details; certificate code	1.0, 2.0, 5.0
B02	EduHub -> Guest	Published catalog; account result; public certificate verification	1.0, 2.0, 5.0
B03	Student -> EduHub	Cart/checkout requests; learning signals; answers; engagement	3.0, 4.0, 5.0, 6.0
B04	EduHub -> Student	Entitlement; receipt; media access; progress; result; certificate	3.0, 4.0, 5.0, 6.0
B05	Instructor -> EduHub	Course/section/quiz drafts; uploads; Q&A response	2.0, 5.0, 6.0
B06	EduHub -> Instructor	Submission/media status; course performance and earnings	2.0, 7.0
B07	Admin -> EduHub	Moderation decision; account/catalog settings; report criteria	7.0
B08	EduHub -> Admin	Approval queue; action status; analytics and revenue reports	7.0
B09	EduHub -> Stripe	Price and order-linked checkout session request	3.0
B10	Stripe -> EduHub	Session reference; signed payment/webhook event	3.0
B11	EduHub -> Blob Storage	Media object upload; protected read and PDF write request	2.0, 4.0, 5.0
B12	Blob Storage -> EduHub	Blob identifiers; upload/retrieval outcome	2.0, 4.0, 5.0

Level 1 / EduHub process decomposition

Seven numbered business transformations, external sources/sinks and ten logical data-store groups.

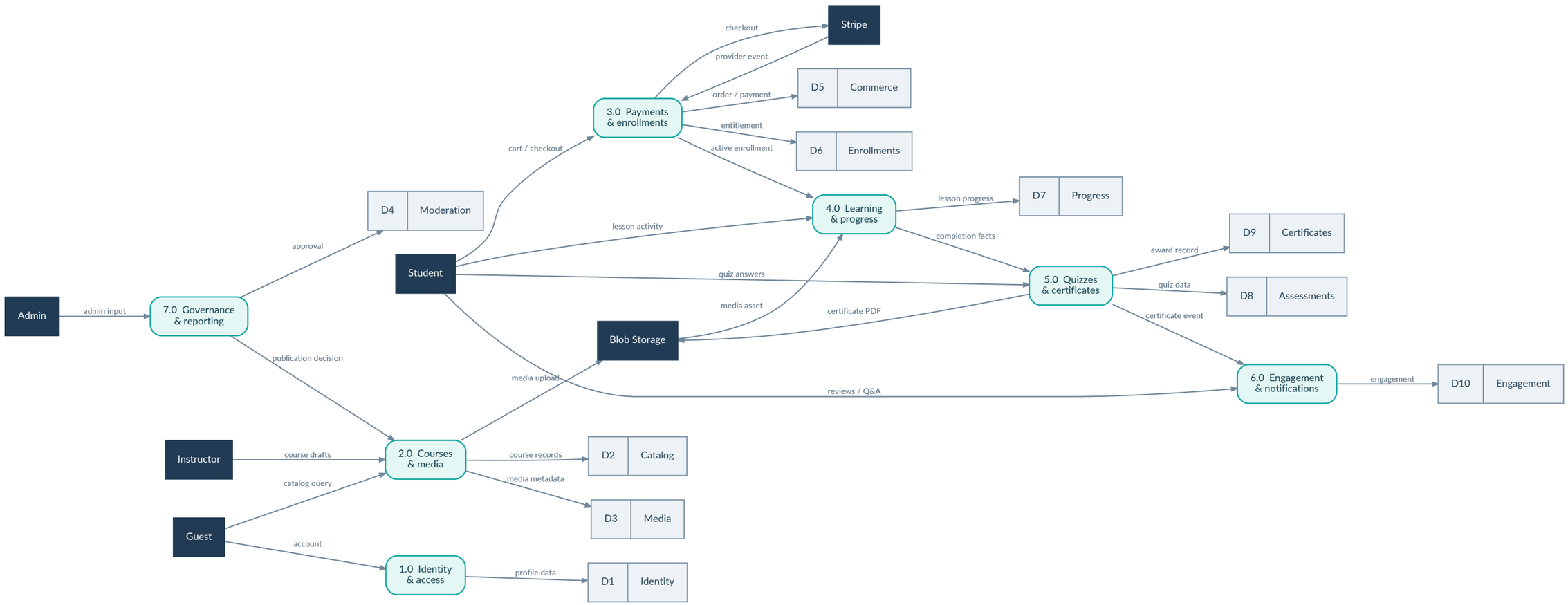


FIGURE 02 / The overview intentionally shows representative cross-process flows; the full external boundary is captured by B01-B12 on page 5 and expanded in the seven Level 2 diagrams. All flows are conceptual, not an API call graph.

Level 1 process dictionary

The seven process boundaries used in all Level 2 expansions

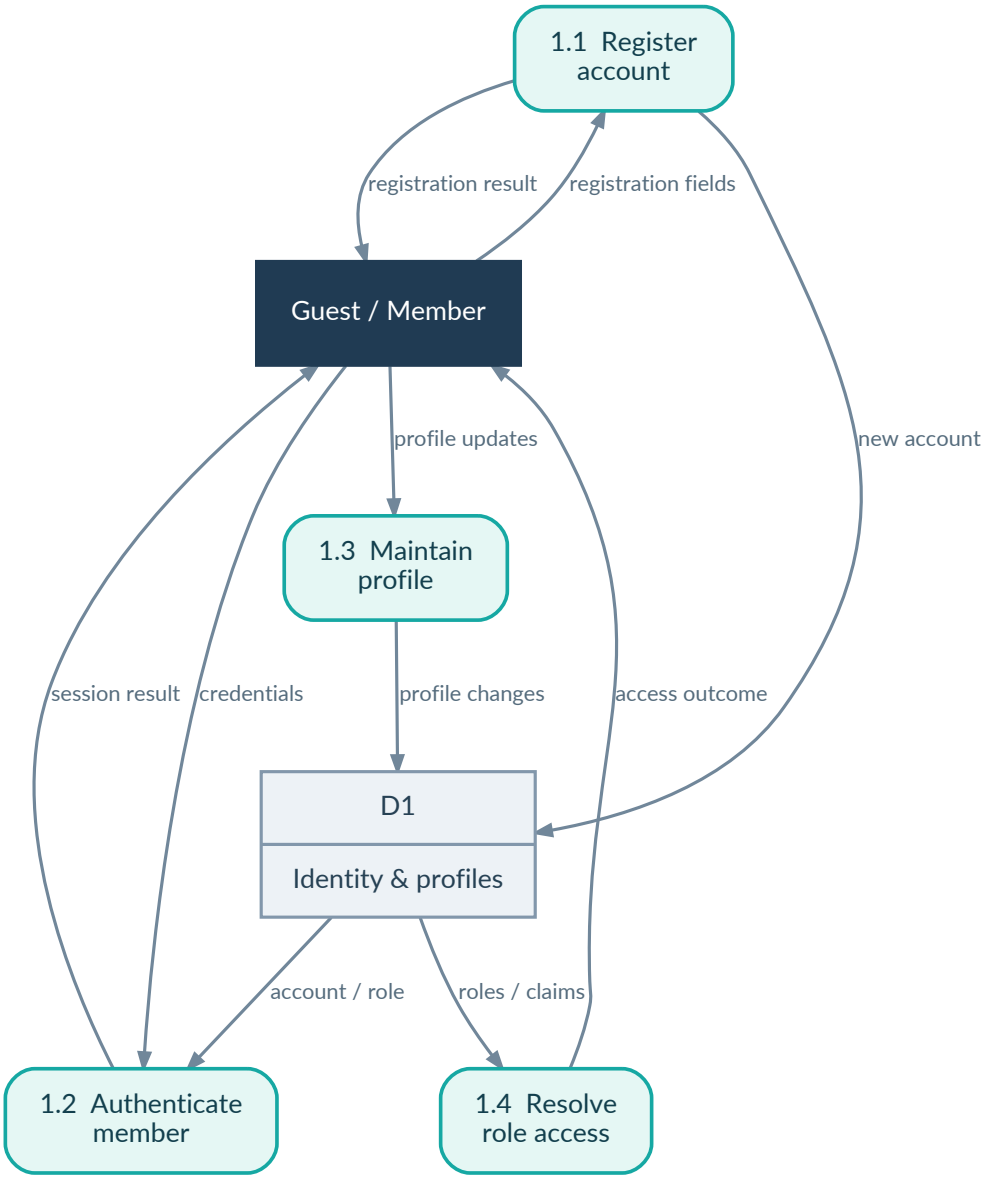
Process responsibilities and the logical stores accessed

ID	TRANSFORMATION	SCOPE / INPUT TO OUTPUT	STORES
1.0	Manage identity and access	Public/member registration, login, profile maintenance, authorization claims.	D1
2.0	Manage courses and media	Catalog, authoring, section order, upload, draft submission and publication visibility.	D2, D3, D4
3.0	Process cart, payment and enrollment	Basket, coupon pricing, Stripe checkout, payment event reconciliation and entitlement.	D5, D6
4.0	Deliver learning and capture progress	Enrollment checks, secure lesson access, playback, notes, bookmarks and completion.	D2, D3, D6, D7
5.0	Assess learning and certify	Quiz definition, scoring, completion eligibility, PDF issue and public verification.	D6, D7, D8, D9
6.0	Manage engagement and notifications	Reviews, helpful votes, Q&A, announcements and event-driven notification records.	D10, D6
7.0	Govern platform and report	Course moderation, user/category administration, settings and aggregated business insight.	D1, D2, D4, D5, D6, D7, D10

Level 2 / Process 1.0

Identity & Access • logical refinement of Level 1 process 1.0

FIGURE 03 / IDENTITY & ACCESS



PROCESS OBJECTIVE

Manage accounts, identity claims and member profile data. Access checks for protected workflows consume these records.

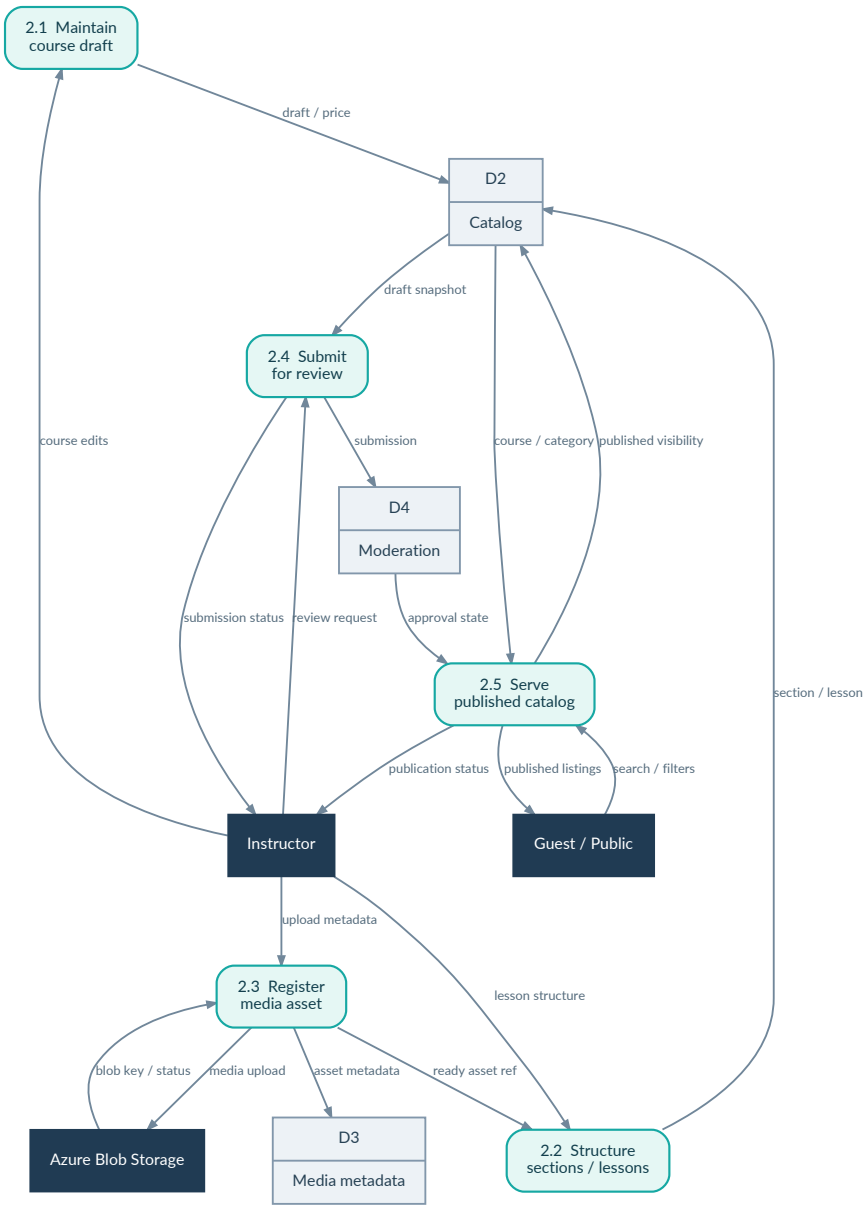
MODEL NOTE

The authorization decision is applied by application services; DFDs model role/identity information, not authentication middleware implementation.

Level 2 / Process 2.0

Courses, Media & Authoring • logical refinement of Level 1 process 2.0

FIGURE 04 / COURSES, MEDIA & AUTHORING



PROCESS OBJECTIVE

Build and publish course content with media references and moderated visibility.

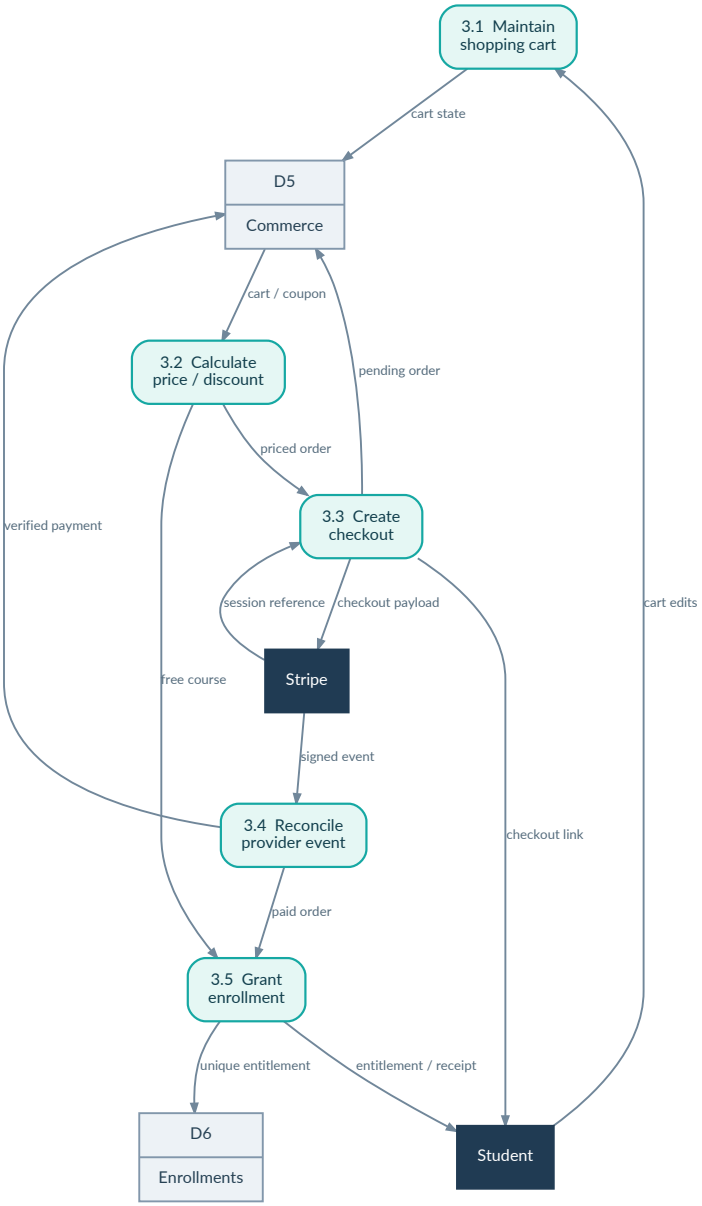
MODEL NOTE

Compression and thumbnails are required by the brief; processing execution is a proposed media workflow. Course publication follows the proposed admin-approval rule.

Level 2 / Process 3.0

Commerce & Enrollment • logical refinement of Level 1 process 3.0

FIGURE 05 / COMMERCE & ENROLLMENT



PROCESS OBJECTIVE

Quote purchases, process Stripe events, and establish one entitlement for each paid or free course.

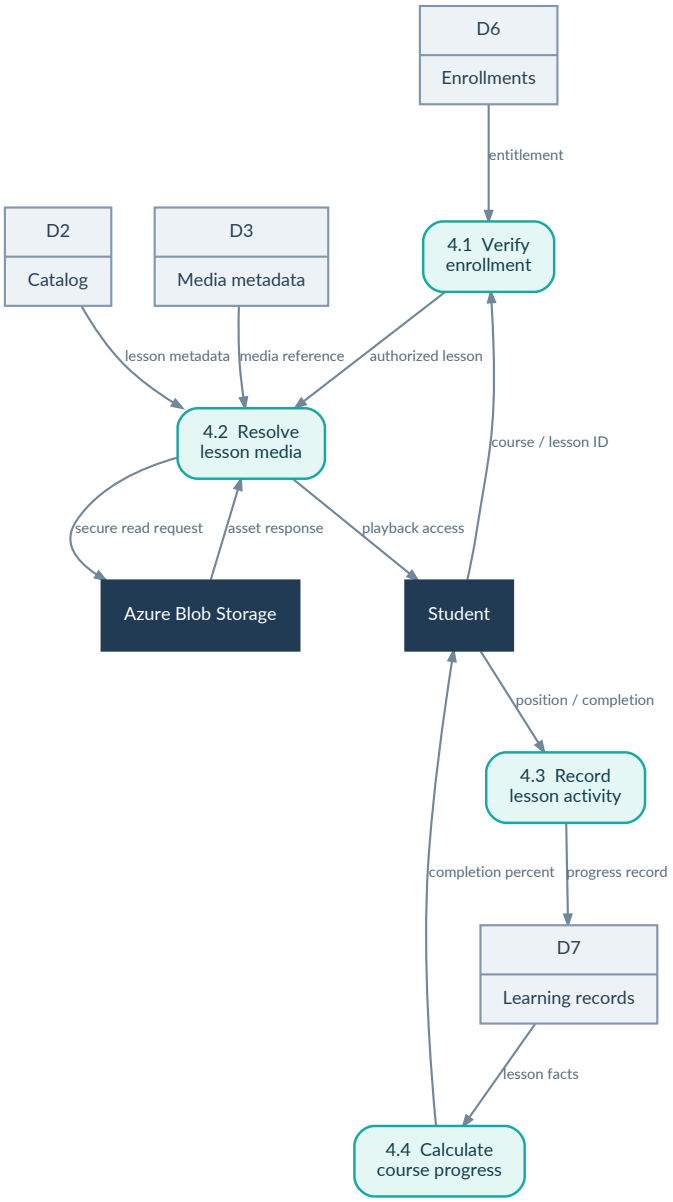
MODEL NOTE

No raw card details enter EduHub. A browser redirect is not payment proof. The free-course path bypasses Stripe but uses the same enrollment constraint.

Level 2 / Process 4.0

Lesson Delivery & Progress • logical refinement of Level 1 process 4.0

FIGURE 06 / LESSON DELIVERY & PROGRESS



PROCESS OBJECTIVE

Provide only authorized learning content and persist resumable lesson-level progress.

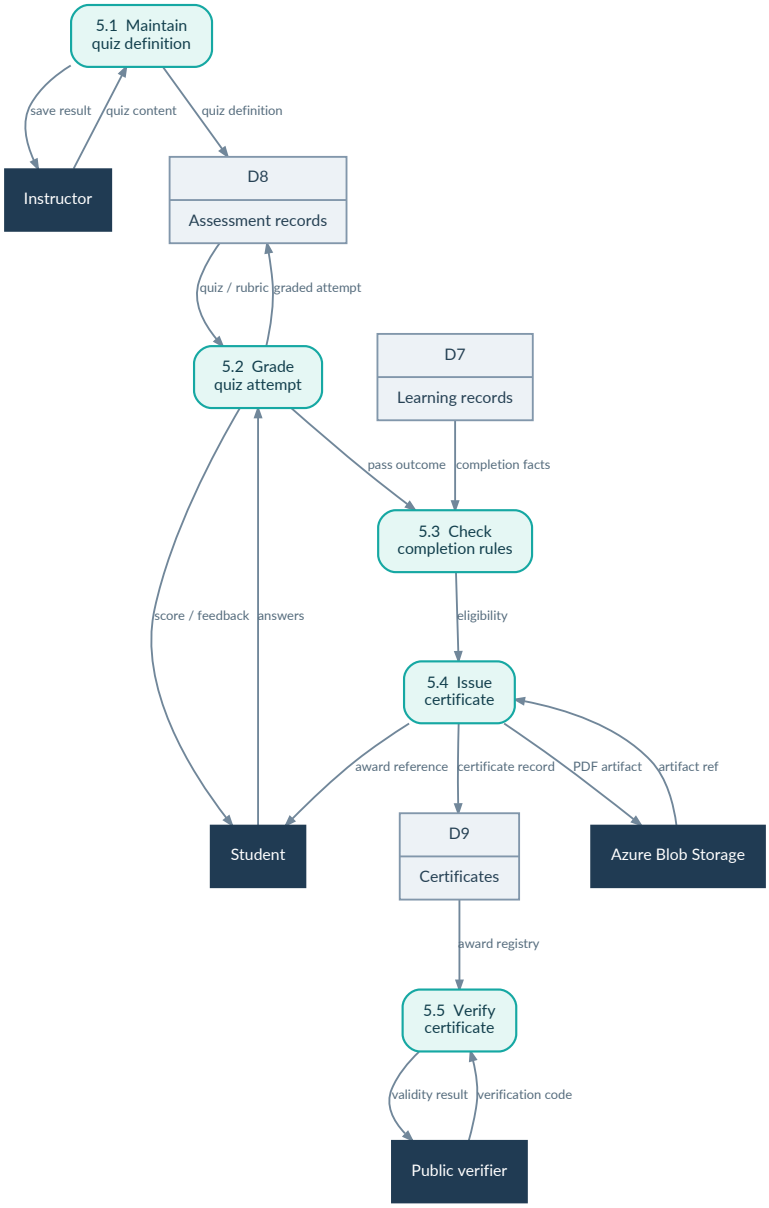
MODEL NOTE

Progress.js and Video.js consume the returned state; these frontend libraries are not data stores. Signed media access is scoped to authorization and expiry.

Level 2 / Process 5.0

Assessment & Certification • logical refinement of Level 1 process 5.0

FIGURE 07 / ASSESSMENT & CERTIFICATION



PROCESS OBJECTIVE

Evaluate quiz submissions, calculate completion eligibility, generate credentials and verify public awards.

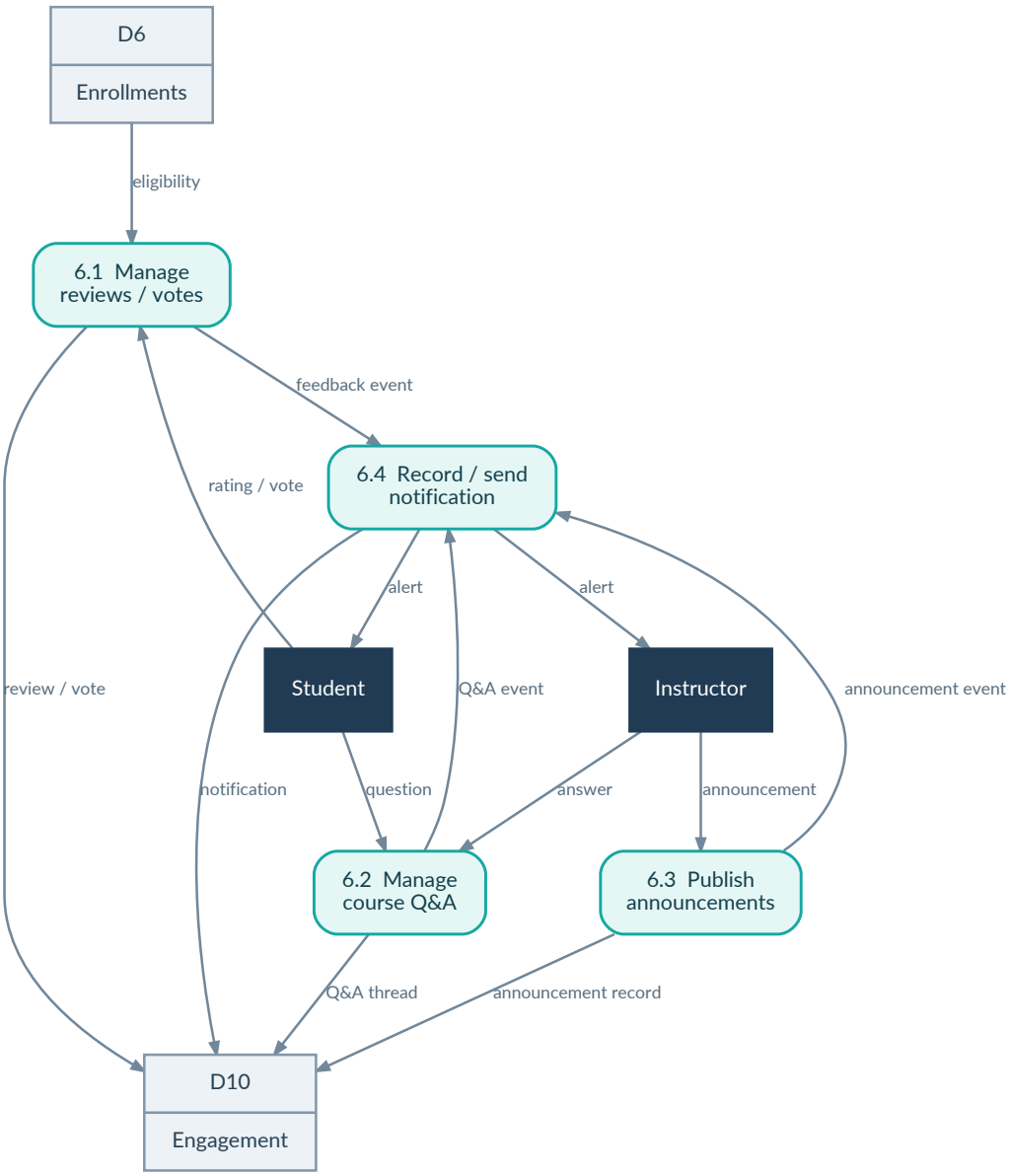
MODEL NOTE

The precise completion formula, retake policy and certificate reissue semantics need team approval; no full financial or identity details appear on public verification.

Level 2 / Process 6.0

Engagement & Notifications • logical refinement of Level 1 process 6.0

FIGURE 08 / ENGAGEMENT & NOTIFICATIONS



PROCESS OBJECTIVE

Persist learner feedback, course Q&A, announcement content, and event-triggered notifications.

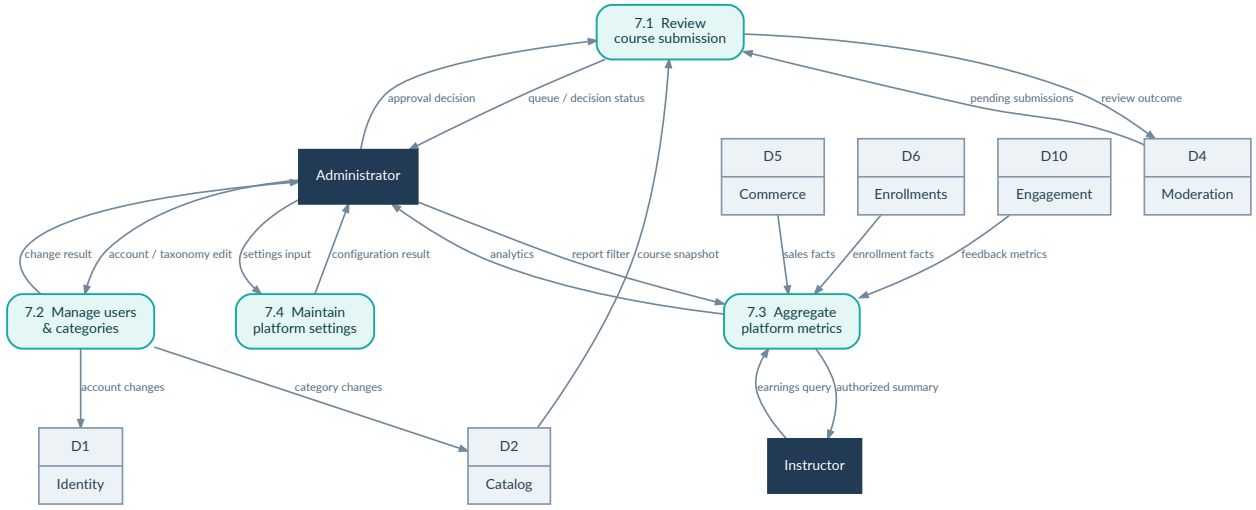
MODEL NOTE

Email and push delivery are not mandated in the supplied brief. An in-app notification record is included as a proposed baseline.

Level 2 / Process 7.0

Governance & Reporting • logical refinement of Level 1 process 7.0

FIGURE 09 / GOVERNANCE & REPORTING



PROCESS OBJECTIVE

Apply administrative decisions and produce reporting data for platform staff and instructors.

MODEL NOTE

The prior 38-entity ERD has no PlatformSetting entity. Settings persistence must be finalized in the later schema revision; the DFD does not pretend a table already exists.

Level 2 flow descriptions

Expanded flow records / part 1 of 4

1.0 / Identity & Access

FLOW ID	NAMED DATA CONTENT	PROCESS OUTPUT / MEANING
F1.1	Registration payload	Creates an ASP.NET Identity account and a role assignment.
F1.2	Authentication result	Returns an authenticated session or denial; never a stored password.
F1.3	Profile changes	Persists allowed student or instructor profile fields.
F1.4	Role-access context	Resolves identity and permissions for protected requests.

2.0 / Courses, Media & Authoring

FLOW ID	NAMED DATA CONTENT	PROCESS OUTPUT / MEANING
F2.1	Course draft	Stores title, description, category, price, level and state.
F2.2	Ordered curriculum	Persists course sections, lessons and their order.
F2.3	Media upload record	Stores secure object reference and processing readiness, not video bytes in SQL.
F2.4	Approval submission	Provides version/visibility information for administrative review.
F2.5	Published catalog	Returns approved published courses and instructor publication status.

Level 2 flow descriptions

Expanded flow records / part 2 of 4

3.0 / Commerce & Enrollment

FLOW ID	NAMED DATA CONTENT	PROCESS OUTPUT / MEANING
F3.1	Cart and coupon data	Persists cart and discount selection; applies Strategy pricing.
F3.2	Checkout session	Creates order-linked checkout; stores provider session reference.
F3.3	Verified payment event	Validates provider signature and deduplicates webhook event.
F3.4	Enrollment entitlement	Grants course access once; paid access only after confirmed payment.

4.0 / Lesson Delivery & Progress

FLOW ID	NAMED DATA CONTENT	PROCESS OUTPUT / MEANING
F4.1	Course access request	Verifies student-course enrollment before protected access.
F4.2	Protected video reference	Resolves entitled lesson and short-lived secure media access.
F4.3	Learning activity	Records playback position and lesson completion flags.
F4.4	Course progress summary	Computes aggregated lesson completion for display and eligibility.

Level 2 flow descriptions

Expanded flow records / part 3 of 4

5.0 / Assessment & Certification

FLOW ID	NAMED DATA CONTENT	PROCESS OUTPUT / MEANING
F5.1	Quiz specification	Stores question types, answers and pass rules.
F5.2	Quiz attempt and result	Scores submission and records answer-level outcomes.
F5.3	Completion eligibility	Combines required lesson and assessment facts.
F5.4	Issued certificate	Creates one unique verification code and PDF reference.
F5.5	Verification response	Returns limited public validity data; no private student records.

6.0 / Engagement & Notifications

FLOW ID	NAMED DATA CONTENT	PROCESS OUTPUT / MEANING
F6.1	Ratings / helpful votes	Enrollment eligibility and uniqueness are checked.
F6.2	Question / answer thread	Persists course-specific discussion.
F6.3	Announcement content	Adds a course notice visible to entitled students.
F6.4	Notification event	Observer event creates recipient-safe in-app notification.

Level 2 flow descriptions

Expanded flow records / part 4 of 4

7.0 / Governance & Reporting

FLOW ID	NAMED DATA CONTENT	PROCESS OUTPUT / MEANING
F7.1	Approval disposition	Records review decision; publication status applied via 2.0.
F7.2	Administrative updates	Maintains user access, categories and tags.
F7.3	Analytics inputs / output	Aggregates enrollment, revenue and engagement; scope differs by role.
F7.4	Platform setting change	Settings behavior required by brief; persistence mechanism unresolved.

Logical data-store dictionary

Ten process-facing stores mapped to the 38 entities in the companion ERD

Logical groups are not physical SQL tables

STORE	BUSINESS LABEL	ENTITY MAPPING (EXACT ERD NAMES)
D1	Identity & profiles	ApplicationUser, ApplicationRole, UserRole, StudentProfile, InstructorProfile
D2	Course catalog	Category, Course, CourseSection, Lesson, Tag, CourseTag
D3	Media asset metadata	MediaAsset
D4	Course moderation	CourseApproval
D5	Commerce records	Cart, CartItem, Order, OrderItem, PaymentTransaction, PaymentWebhookEvent, Coupon, CouponRedemption, InstructorEarning

Persistence boundary

D3 stores media metadata and Azure Blob keys - not binary video content. D5 stores order and payment events, not Stripe cardholder data.

Logical data-store dictionary

Part 2 of 2 / Enrollment, learning, assessment, certificates and engagement

STORE	BUSINESS LABEL	ENTITY MAPPING (EXACT ERD NAMES)
D6	Enrollment records	Enrollment
D7	Learning records	LessonProgress, LessonNote, LessonBookmark
D8	Assessment records	Quiz, QuizQuestion, QuizOption, QuizAttempt, QuizAnswer
D9	Certificate registry	Certificate
D10	Engagement records	CourseReview, ReviewHelpfulVote, CourseQuestion, CourseAnswer, CourseAnnouncement, Notification

Schema alignment

The ten logical stores jointly map to 38 previously identified ERD entities. Administrative platform settings remain an open persistence decision, not an invented 39th table.

Cross-process data exchange

The named information links that connect the Level 1 transformations

Cross-process data dictionary

ID	SOURCE -> TARGET	INFORMATION CARRIED	REASON / CONSUMING FUNCTION
X01	3.0 -> 4.0	Active enrollment / paid entitlement	Only successful payment or valid free enrollment permits lesson access.
X02	4.0 -> 5.0	Completion facts / course progress	Assessment/certification consumes learning records and completion summary.
X03	5.0 -> 6.0	Pass or issued-certificate event	An event can create a learner notification; details are access-scoped.
X04	7.0 -> 2.0	Course approval / rejection	Publication visibility is updated from recorded moderation decision.
X05	2.0 -> 4.0	Published curriculum and media metadata	Only approved published lessons are eligible for student playback.
X06	2.0 -> 5.0	Quiz definition and attachment	Instructor changes quiz configuration referenced by assessment logic.
X07	3.0 -> 7.0	Order/revenue and enrollment facts	Admin/instructor reporting uses role-filtered aggregate data.
X08	6.0 -> 7.0	Engagement signals	Review, Q&A and notification activity contribute to reports.
X09	1.0 -> all protected processes	Principal / role scope	Authorization is evaluated for student, instructor and admin functions.

Balancing & traceability

Parent-level inputs and outputs accounted for in the Level 1 / 2 refinements

Context-to-Level-1 traceability

EXTERNAL ENTITY	BOUNDARY FLOWS	CONSERVED LEVEL 1 EXCHANGE
Guest	B01/B02	1.0 account; 2.0 catalog; 5.0 public verification
Student	B03/B04	3.0 commerce; 4.0 lessons; 5.0 quizzes/awards; 6.0 feedback
Instructor	B05/B06	2.0 course/media; 5.0 quizzes; 6.0 announcements; 7.0 insights
Admin	B07/B08	7.0 governance, categories, users, reporting
Stripe	B09/B10	3.0 checkout session and signed payment events
Azure Blob	B11/B12	2.0 uploads; 4.0 protected reads; 5.0 issued PDF

Level 1 to Level 2 decomposition check

PARENT	CHILD PROCESSES	BOUNDARY CHECKS
1.0	1.1, 1.2, 1.3, 1.4	External source/sink + named process-store flows
2.0	2.1, 2.2, 2.3, 2.4, 2.5	External source/sink + named process-store flows
3.0	3.1, 3.2, 3.3, 3.4, 3.5	External source/sink + named process-store flows
4.0	4.1, 4.2, 4.3, 4.4	External source/sink + named process-store flows
5.0	5.1, 5.2, 5.3, 5.4, 5.5	External source/sink + named process-store flows
6.0	6.1, 6.2, 6.3, 6.4	External source/sink + named process-store flows
7.0	7.1, 7.2, 7.3, 7.4	External source/sink + named process-store flows

Use case & requirement coverage

Coverage based on the previously cataloged 48 EduHub use cases

Process-to-use-case relationships

PROCESS	BUSINESS FUNCTION	USE CASE IDS
1.0	Manage identity and access	UC-01, 02, 06
2.0	Manage courses and media	UC-03, 04, 25-32, 46
3.0	Process cart, payment and enrollment	UC-07-11, 43, 44
4.0	Deliver learning and capture progress	UC-12-15, 17-18, 47
5.0	Assess learning and certify	UC-05, 16, 23-24, 33, 45
6.0	Manage engagement and notifications	UC-19-22, 35-36, 48
7.0	Govern platform and report	UC-34, 37-42

Coverage interpretation

Use cases can touch multiple business processes. The process IDs serve as cross-document anchors; exact FR IDs should be validated against the requirements PDF before final sign-off.

Example end-to-end coverage

Paid course entitlement	UC-08 / 09 / 10 / 43 / 44 -> 3.0; UC-12 / 13 -> 4.0
Course approval	UC-25 / 31 -> 2.0; UC-37 / 38 -> 7.0; publication -> 2.0
Certificate issuance	UC-14 / 16 -> 4.0 / 5.0; UC-45 -> 5.0; UC-05 / 23 -> 5.0

Trust boundaries & data minimization

Where data enters from users and external providers - and what must be validated

Sensitive flows and processing controls

AREA	DFD BOUNDARY / ROUTE	PROPOSED INTEGRITY / PRIVACY CONTROL
Identity	Guest/member credentials -> 1.0 -> D1	Identity hashing, secure session, RBAC; avoid credentials in analytics/logs.
Checkout	3.0 -> Stripe -> 3.0 -> D5/D6	Server-side session creation; validate Stripe signature and event uniqueness.
Media	Instructor -> 2.0 -> Azure Blob / D3	Restrict upload identity; protected blob; short-lived authorized access.
Learning	Student -> 4.0 -> D7	Verify enrollment and resource ownership before reading or updating progress.
Quiz	Student -> 5.0 -> D8	Keep correct answers out of pre-submit responses and enforce grading rules server-side.
Certificate	5.0 -> D9/Blob; Public -> 5.5	Unique code; public verification returns only non-sensitive credential metadata.
Reporting	D5/D6/D10 -> 7.0 -> Admin/Instructor	Aggregate role-scoped metrics; instructors must not see other instructors data.

External system representation

Stripe and Azure Blob are external entities at the logical DFD boundary. They are not modeled as EduHub SQL tables; EduHub persists payment references and media metadata locally.

DFD quality & review checklist

Reviewer-ready validation of modeling rules, ambiguity and decision ownership

Model quality audit

PASS	One process only on Context / Level 0; no internal store on that view.
PASS	Seven top-level transformations carry hierarchical identifiers 1.0 to 7.0.
PASS	All six context external actors reappear in the Level 1 ownership mapping.
PASS	No intentional direct external-entity-to-store or store-to-store flow.
PASS	Ten DFD logical stores map onto the established 38 ERD entities.
REVIEW	Publish-before/after-admin-approval behavior requires team confirmation.
REVIEW	Completion and quiz retake policy requires a signed business rule.
REVIEW	PlatformSetting persistence is not represented by the current 38-entity schema.
REVIEW	Background media conversion and notification delivery implementation is proposed.

Supervisor / team sign-off decisions

A. Define administrative approval before first publication and after material edits.

B. Approve course completion formula and required section-level / course-level quizzes.

C. Confirm whether certificates and settings use SQL metadata plus Azure artifacts.

D. Approve logical store mapping if the data schema changes before implementation.

References & assumptions register

Traceable source provenance and formal status of this DFD package

Project-provided primary sources

[P1] EduHub - Online Learning Platform. Project 3 brief, 5 pages, supplied with this conversation.

Primary feature and technology scope.

[P2] EduHub Software Architecture. Companion design deliverable.

Logical layers and external integrations.

[P3] EduHub Entity-Relationship Diagram and Logical & Physical Schema.

38 entity names and relational modeling baseline.

[P4] EduHub Use Case Diagram & Descriptions.

48 use-case inventory and external actors.

[P5] EduHub Functional & Non-Functional Requirements.

Derived scope, acceptance criteria and open decisions.

DFD notation and integration references

[R1] IBM

What Is a Data Flow Diagram?
https://www.ibm.com/think/topics/data-flow-diagram

[R2] IRS

Analysis Techniques and Deliverables - DFD elements and leveling
https://www.irs.gov/irm/part2/irm_02-005-011

[R3] Stripe

Webhook endpoints and signature verification
https://docs.stripe.com/webhooks

[R4] Microsoft

Microsoft Learn - Azure Blob Storage introduction
https://learn.microsoft.com/en-us/azure/storage/blobs/storage-blobs-introduction